

Implementation Matters: Generalizing Treatment Effects in Education

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Why and when does targeted instruction "work"?

- Targeted instruction is e.g. "teaching at the right level" (TaRL).
- Popularised and evaluated in developing contexts by Banerjee, Duflo and others – but treatment effects vary by an order of magnitude (0.07 to 0.78 SDs).
- We aggregate evidence across studies and find that accounting for differences in implementation can uncover generalizable effects.
- We show that implementation can be exogenously changed via a new RCT in Botswana – and this improves learning.
- We provide new models to account for uncertainty in implementation jointly with treatment effects. Results show large unrealised gains from better implementation.

Targeted instruction is popular now in developing settings

- Targeted instruction is a type of intervention that:
 - ① assesses student level/attainment,
 - ② splits the students up by that level,
 - ③ teaches material directly targeted to that level.
- The most famous of these programs is "Teaching at the Right Level" (TaRL), developed by Pratham in India.
- Various different targeted instruction approaches are actively being adopted by dozens of countries and scaled up to over 60 million children worldwide.
- Efforts involve World Bank, NGOs & governments.

We consider studies that monitored their implementation

- Targeted Instruction requires complex implementation:
 - a student attendance
 - b teacher attendance
 - c materials usage
 - d student level assessment
 - e actual targeted delivery...
- We consider studies which sent surveyors to monitor the intervention and to measure (a-c). 3 interventions additionally reported (d-e).
- We follow the existing studies in defining "take-up" or "compliance" as (a-c) occurring, and this defines our TOT.
- Items (d-e) we call "fidelity"; we will address this later.

Our Data & Approach

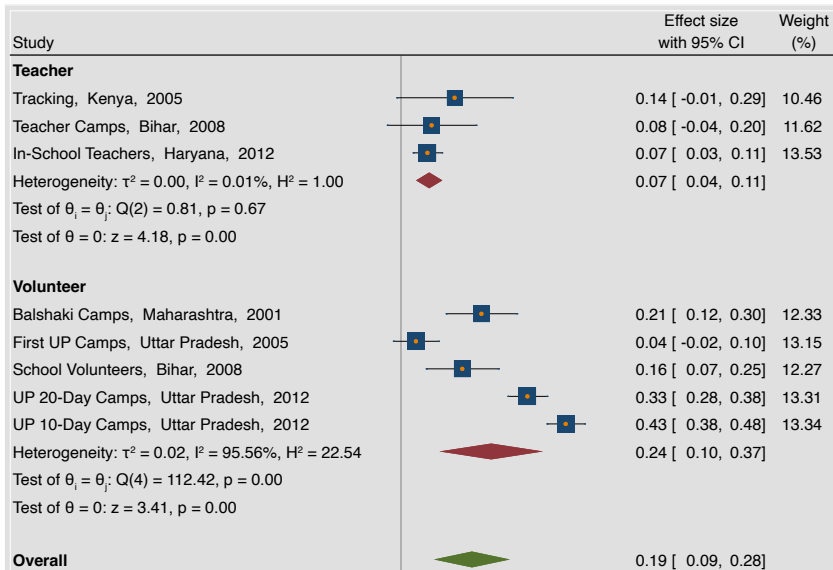
- We use microdata from 8 treatment arms across 5 contexts: Banerjee et al. (2007) in Maharashtra, Duflo et al (2011) in Kenya, Banerjee et al. (2017) in Bihar, Uttar Pradesh, and Haryana.
- Programs differ in baseline student attainment, takeup rate among students (from 8% to 90%), taught by teachers or volunteers.
- Main outcome: average literacy and numeracy from a test like ASER.
- First we consider intention-to-treat (ITT) and treatment-on-treated (TOT) separately.
- Start with frequentist random-effects meta-analysis and progress from there.

Our microdata covers 73,500 students

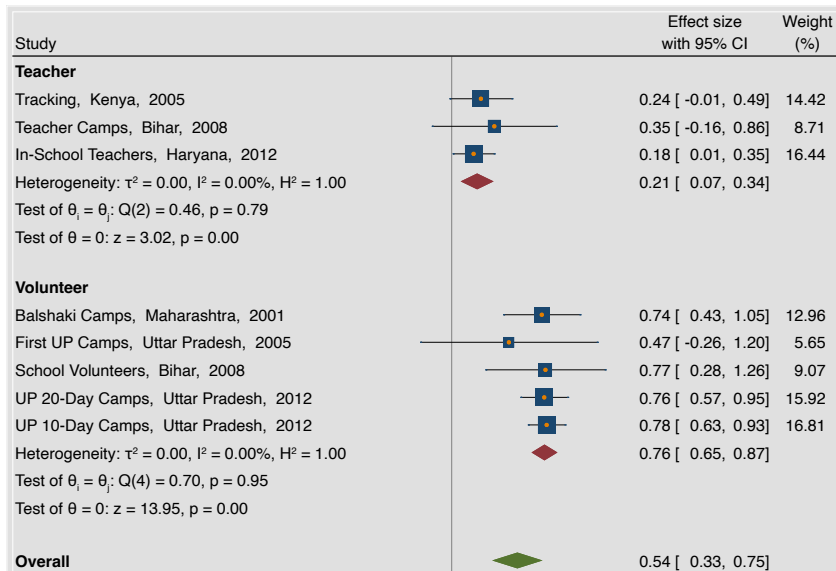
Table: Studies Considered for Evidence Aggregation

Authors	State/Country	Treatment Arm	Delivery	Sample Size
<i>Studies included</i>				
Banerjee et al. (2007)	Maharashtra, India	Balshaki Camps	Volunteer	10000
Banerjee et al. (2010)	Uttar Pradesh, India	First UP Camps	Volunteer	9442
Duflo et al (2011)	Kenya	Tracking	Teachers	6000
Banerjee et al. (2017)	Bihar, India	School Volunteers	Volunteer	3325
Banerjee et al. (2017)	Bihar, India	Teacher Camps	Teachers	2474
Banerjee et al. (2017)	Uttar Pradesh, India	UP 10-day Camps	Volunteer	17266
Banerjee et al. (2017)	Uttar Pradesh, India	UP 20-day Camps	Volunteer	13054
Banerjee et al. (2017)	Haryana, India	In-school Teachers	Teachers	11966
Total	5	8	-	73527

ITTs: big effects but heterogeneous for volunteers



TOTs: volunteer effects now larger + generalise



When do these correlations matter?

- We found: Program takeup and delivery model are more important than country context or baseline attainment.
- Suggests implementation factors drive differences in results.
- Problem 1: Is the relationship causal? Can implementation even be changed in practice?
 - Solution: new RCT in Botswana to vary implementation.
- Problem 2: Take-up and other implementation metrics are uncertain and that uncertainty should be in the aggregation procedure.
 - Solution: new joint model of uncertainty on all variables.

New RCT in Botswana: Implementation can be changed

- It is possible that implementation is a fixed feature of an environment and not a policy lever in itself. (ie perhaps the NGO is always just trying its best...)
- Even if it can be changed, perhaps omitted variables drive both implementation and TEs, and effect above is not causal?
- We now show proof of concept that the causal impact of implementation can be studied rigorously because it can be varied systematically.
- New RCT with Youth Impact in Botswana: treatment group sessions increased the degree of targeting/tailoring to student abilities.

Botswana Results: Increased Learning Outcomes

	Learning Outcomes: Number Recognition			
	(1)	(2)	(3)	(4)
Subgrouping by Similar Sub-Levels	0.190 (0.098) [0.058]	0.225 (0.099) [0.027]	0.211 (0.087) [0.019]	0.208 (0.084) [0.017]
Observations	1069	1069	1069	1069
Baseline Level Control	Yes	No	Yes	Yes
Baseline # Recognition Control	No	Yes	Yes	Yes
Region Fixed Effects	No	No	No	Yes

Notes: All standard errors are robust and clustered at the class level. P-values are reported in brackets. Learning gains are expressed in terms of standard deviations using the control group standard deviation.

New joint model of effects & implementation

- If implementation can be varied, then it is a random variable that may covary with the potential treatment effects themselves.
- (We claim) It also has a specific relationship to observed impact: it dials the realised treatment effect up or down – ie. multiplicative.
- Both IV and meta-regression condition on the measured values of these covariates, without accounting for uncertainty in their distribution.
- Jointly aggregating the information on the effects and the implementation is possible with a new model.

Extending from "compliance" π_c to "implementation" m

- Conventional analysis uses this structure:

$$ITT = ITT_c \pi_c$$

- We can shift from π_c as a proportion of types to instead being a proportion or extent to which the program is implemented. Call it $m \in [0, 1]$. (And jettison the notion of "types" now.)
- The object that was called " ITT_c ", or TOT, is now the effect we would see if the program was "fully" implemented $m = 1$.
- The effect we do see, the " ITT ", is scaled up/down by m .
- The identifying structure is that there is a single "latent effect" we access fractionally.

Latent Effect Aggregation

- Formally, we consider a set of program contexts indexed by $j = 1, 2, 3 \dots J$ and define three relevant objects.
- The implementation factor (m-factor), denoted $m_j \in [0, 1]$ for a setting j , is the extent or proportion to which the program was effectively implemented in setting j .
- The Latent Treatment Effect LTE_j , denoted $\theta_j \in \mathbb{R}$ for a setting j , is the impact achievable when the program is fully implemented.
- The Realized Treatment Effect RTE_j is the observed impact of the program in setting j , defined as:

$$RTE_j \equiv m_j \theta_j$$

Latent Effects can be identified if m is measured

- Latent Treatment Effects matter just as much if not more than Realised Treatment Effects – especially for generalizability to future interventions.
- If implementation m is not measured (with error is fine), the Latent TEs are not identified from the Realised TEs and nor is their variation.
- We build and fit a Bayesian hierarchical model that decomposes each RTE_j into m_j and LTE_j while aggregating.
- We consider takeup and program fidelity as m_{1j}, m_{2j} .

Average Latent ITT of TaRL approach TOT

Table: Model with Takeup as Implementation factor: Posterior Distribution on Effects

	mean	2.5%	25%	50%	75%	97.5%	Rhat
<i>Panel A: Latent Treatment Effects (All)</i>							
Hypermean	0.418	0.231	0.361	0.414	0.473	0.616	1.002
HyperSD	0.207	0.068	0.136	0.188	0.256	0.464	1.002
<i>Panel B: Latent Treatment Effects (Teacher)</i>							
Hypermean	0.239	-0.104	0.154	0.223	0.305	0.697	1.031
HyperSD	0.235	0.005	0.060	0.142	0.312	0.922	1.021
<i>Panel C: Latent Treatment Effects (Volunteer)</i>							
Hypermean	0.486	0.166	0.420	0.474	0.554	0.809	1.012
HyperSD	0.233	0.017	0.087	0.164	0.296	0.930	1.006

Note: This inference is generated by $J = 7$ studies.

Average Latent TOT matches largest effects in literature

Table: Model with Fidelity and Takeup as Implementation Factors: Posterior Distributions of Effects

	mean	2.5%	25%	50%	75%	97.5%	Rhat
<i>Panel A: Fidelity on TOT</i>							
Hypermean	0.846	0.379	0.734	0.834	0.935	1.410	1.011
HyperSD	0.392	0.008	0.091	0.220	0.512	1.673	1.013
<i>Panel B: Fidelity and Takeup Jointly</i>							
Hypermean	1.200	0.126	0.998	1.140	1.356	2.324	1.009
HyperSD	0.782	0.012	0.133	0.368	0.949	4.304	1.014

Note: This inference is generated by $J = 3$.

Conclusion: We must evaluate implementation itself

- Implementation is not a fixed property of an environment but a variable that can be studied and changed.
- In the context of targeted instruction, implementation almost entirely explains heterogeneity in effects across contexts.
- Cross-study evidence is correlational but new RCTs can be run to study these correlations – we provide a blueprint we hope will be adopted for this.
- Without gathering and analysing this information as part of policy evaluation, *we are mostly guessing as to why things succeed.*

Literacy Example

Letter

b t g
f u
n d
v m r

Word

mother after
school dog
banana sorry
doll please river
cat

Story

Boago and Pearl were friends. They liked to play together. They found sweets in the market. Boago bought sweets. There were not enough to share with Pearl. Pearl was sad and started crying. She didn't want to play with Boago. Boago said sorry. They became good friends again.

1. What did Boago buy?
2. Why did Pearl start crying?

Para

Thato is not feeling well.
She is in pain.
She went to the clinic.
A doctor helped her.

Numeracy Example

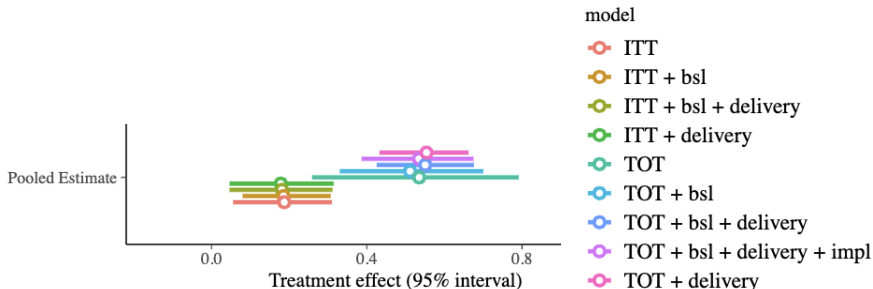
$\begin{array}{r} 62 \\ + 18 \\ \hline \end{array}$	$\begin{array}{r} 33 \\ + 49 \\ \hline \end{array}$	$\begin{array}{r} 16 \\ + 47 \\ \hline \end{array}$
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$\begin{array}{r} 91 \\ - 52 \\ \hline \end{array}$	$\begin{array}{r} 42 \\ - 38 \\ \hline \end{array}$	$\begin{array}{r} 81 \\ - 43 \\ \hline \end{array}$
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$\begin{array}{r} 26 \\ \times 3 \\ \hline \end{array}$	$\begin{array}{r} 38 \\ \times 2 \\ \hline \end{array}$	$\begin{array}{r} 12 \\ \times 5 \\ \hline \end{array}$
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$\begin{array}{r} 6 \overline{)93} \end{array}$	$\begin{array}{r} 4 \overline{)53} \end{array}$	$\begin{array}{r} 3 \overline{)49} \end{array}$
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Delivery model matters more than baseline scores



All regression coefficients are heavily regularised to zero to avoid overfitting.

Hierarchical Implementation Model

Hierarchical likelihood based on Rubin (1981):

$$\widehat{RTE}_j \sim N(m_j \theta_j, \hat{se}_j^2)$$

$$\hat{m}_j \sim N(m_j, \hat{se}_{m_j}^2)$$

$$\theta_j \sim N(\theta, \sigma_\theta^2)$$

The parameter θ is called the Hypermean and the σ_θ^2 is the Hypervariance.

Add priors on $\theta, \sigma_\theta^2, \hat{se}_j^2, \hat{se}_{m_j}^2$ to go full Bayes.

Also possible to aggregate m_j – shrink towards some average m – if desired.